

Larry Nguyen

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Education

University of Maryland, College Park, MD

Anticipated Graduation: Dec. 2027

B.S. in Electrical Engineering

Relevant Coursework: Advanced Analog and Digital Electronics, Digital Logic Design, Electric Circuits, Electromagnetics, Introduction to Micro and Nanoelectronics, Signal & System Theory

Experience

Pho An & Grill – Silver Spring, MD

Jun. 2025 – Present

Server & Busser

- Served customers, handled food service, and helped maintain smooth operations during busy shifts
- Trained new employees on restaurant procedures and day-to-day service

Navy Yard & UMBC Mechanical Rooms — Washington, D.C. / Baltimore, MD

Sept. 2023 – May 2026

Externship | Boiler and Chiller Operations

- Shadowed boiler/chiller operators across two facilities: contactor replacement, wire pulling, and OV diagnostics
- Observed VFDs, relays, and contactors in mechanical cooling/heating equipment using multimeters and non-contact voltage testers

Projects

Adjustable Linear Bench Power Supply

Jul. 2026 – Present

Personal Project

- Designed a 0–15V, 1.2A linear power supply with rectification, filtering, relay protection, and regulation stages
- Simulated in LTspice before layout; completed PCB layout and fabrication
- Sized the pass-transistor heatsink from a worst-case ~28W dissipation calculation and designed current-limiting protection via sense resistor
- Diagnosed and resolved a hard fault on regulator output by tracing the failure through the regulation chain

DC-DC Buck Converter with PID Control

Jun. 2026 – Jul. 2026

Personal Collaborative Project

- Designed a 12V-to-5V, 2A synchronous buck converter with a secondary 3.3V MCU-compatible voltage divider
- Selected gate drivers, MOSFETs, and RLC components for 101 kHz switching operation
- Validated switching waveform and ripple on an oscilloscope against simulation
- Diagnosed a feedback-resistor footprint mismatch (3.21V vs. 3.3V) and corrected it with an off-board mount

Crushin' It — ENES 100

Sept. 2023 – Dec. 2023

Semester Project

- Collaborated on a six-person team to design an autonomous vehicle that located, measured, and reached a target
- Led electronics for sensors, actuators, and microcontroller control; awarded Most Creative OTV among six teams
- Wired color and Time-of-Flight sensors and three servos to an Arduino Mega, routing I²C and PWM signals

Arcade Machine Restoration & Modernization Project

Dec. 2025

Personal Project

- Upgraded a vintage arcade cabinet to support 60,000+ retro games via a Raspberry Pi 5 running Batocera Linux
- Removed obsolete hardware and rewired 15 LED-illuminated buttons/joysticks to a USB encoder board on the Pi
- Wired the Raspberry Pi to the monitor's controller board over HDMI, replacing the cabinet's stock video path

Skills

Hardware: Arduino, Basic Electrical Equipment, Circuit Design, Electronics, KiCad, LTspice, Schematics, Soldering

Software: Autodesk CAD, Excel, GitHub, MATLAB, MIPS Assembly, Verilog